EK-13 RAG Retrieval, Chunking and Citation Guide

Document ID: EK-13 | Category C: Evaluation knowledge base

1. Document metadata

Field	Value
Document ID	EK-13
Filename	EK-13_RAG_Retrieval_and_Citation_Guide.pdf
Category	C - Evaluation knowledge base
Version	1.0
Creation date	2026-10-14
Data status	Factual technical guidance with computed metric examples
Main topics	RAG; Chunking; Hybrid retrieval; Metrics
Relevant skills and technologies	RAG pipeline; Vector search
Relevant entity IDs	EVA-005, TEC-004, EK-07

Purpose and scope: specify chunking, metadata, hybrid retrieval, query routing, the answer contract and retrieval metrics for the proposal-review assistant.

2. Executive summary

Chunk by heading with entity IDs, filter by metadata, fuse semantic and keyword results, require citations and measure Recall@K, MRR and nDCG@K.

3. Chunking rules

- Chunk by heading; one main topic per chunk (about 150 to 400 words).
- Repeat the entity ID (JOB, PRP, FRE, PRJ) in every section heading and table caption.
- Keep tables small enough to be understandable as text.
- Do not split a table row across chunks.

4. Chunk metadata schema

Field	Example	Use
document_id	PRP-007	Source attribution
pdf_id	FP-02	File reference
category	Freelancer proposals	Filtering by category
job_id	JOB-004	Restrict to one job
proposal_id	PRP-012	Restrict to one proposal
freelancer_id	FRE-005	Restrict to one freelancer
skills	Docker;Kubernetes	Skill filter

5. Hybrid retrieval

Run semantic (vector) search and keyword search (for example BM25) and fuse the ranked lists with reciprocal rank fusion: score(d) = sum over lists of 1 / (60 + rank). Example: list 1 = A, B, C and list 2 = B, C, A give A 0.03227, B 0.03252, C 0.03200; B ranks first. Apply metadata filters before searching.

6. Query routing examples

Question type	Filters	Sources to retrieve
Why did PRP-012 get its score?	proposal_id=PRP-012	Stored score, proposal text, portfolio of FRE-005, EVA methods
Who has Spring Boot and PostgreSQL experience?	skills	PORT-001, projects
What does JOB-003 require?	job_id=JOB-003	CR-01, CR-03, CR-06
How is budget scored?	category=Evaluation	EVA-002, EK-05

7. Answer contract for the language model

- Quote stored scores and database facts.
- Cite every factual claim.
- If no evidence is retrieved, say so.
- Separate facts from interpretation.
- Never invent history, ratings, prices or scores.

8. Retrieval metrics with a worked example

Relevant documents ['A', 'B']; retrieved ranking ['C', 'A', 'D', 'B', 'E']. Recall@3 = 0.50; Recall@5 = 1.00; MRR (reciprocal rank of the first relevant document) = 0.50; DCG@5 = 1.062; ideal DCG = 1.631; nDCG@5 = 0.651. Use retrieval_test_questions.csv to compute these values over all questions.

9. Failure handling

- No relevant chunk: answer 'no evidence found' and name what was searched.
- Conflicting chunks: prefer the case study over the profile and report the conflict.
- Chunks from the wrong entity: drop them.

10. Summary of key findings

- Metadata filters narrow retrieval to the right entity.
- RRF fuses semantic and keyword rankings.
- Metrics are computed on a test set.

11. Keywords and alternative terminology

RAG; chunking; metadata; hybrid retrieval; reciprocal rank fusion; Recall@K; MRR; nDCG; query routing; citation